RainCastSG: Singapore Rainfall Analysis & Prediction

Exploring Spatiotemporal Patterns & 30-Minute Threshold Exceedance with Apache Spark

DSA5208 - Scalable Distributed Computing for Data Science | National University of Singapore

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Project Team Members

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Executive Summary

This project presents RainCastSG, an end-to-end scalable distributed computing pipeline built with Apache Spark (PySpark 4.0.4) to analyze eight years (2017-2024) of 5-minute rainfall observations from monitoring stations across Singapore and predict short-horizon extreme rainfall accumulation. In Task 1, we conduct extensive exploratory data analysis on 46.8 million observations from data.gov.sg (collection 2279), characterizing interannual shifts (e.g., dry 2019 vs. wet 2021/2022), strong seasonal monsoonal cycles, a pronounced afternoon convective peak (15:00-16:00 SGT), and localized spatial rainfall gradients. In Task 2, we formulate a probabilistic binary classification framework estimating P(R_{t->t+30m} > x | station, time, history) for rainfall thresholds x in [0, 10] mm. Using a Spark MLlib Logistic Regression pipeline evaluated chronologically on held-out 2024 data, the model achieves a pooled ROC-AUC of 0.9063 and PR-AUC of 0.4076. Furthermore, Platt-scaling probability calibration fitted on 2023 validation data reduces Brier score error by 11.0% to 30.9% across all evaluated thresholds relative to an historical-frequency baseline, with zero probability monotonicity violations across 633,357 test observations. Complete source code, trained models, and reproduction CLI are provided.

1. Introduction & Task Objectives

Singapore experiences a tropical rainforest climate (Koppen Af) characterized by high ambient temperatures, high humidity, and abundant precipitation throughout the year. Precipitation is predominantly driven by localized mesoscale convective storms and seasonal monsoon surges. High-frequency automated meteorological stations recorded at 5-minute intervals provide a valuable empirical record to examine weather phenomena and build predictive tools for urban flood management, stormwater routing, and public safety.

The project is structured around two core distributed data science tasks:

- Task 1 - Weather Pattern Exploration: Analyze the historical rainfall data across annual, monthly, diurnal, intensity, and spatial dimensions from 2017 to 2024, uncovering fundamental meteorological patterns and interannual variability using scalable Spark SQL aggregations.

- Task 2 - Probabilistic Rainfall Prediction: Construct an Apache Spark MLlib machine learning pipeline that estimates the probability that a designated monitoring station will accumulate more than a user-specified rainfall threshold x (in mm) over the subsequent 30 minutes, conditioning on the station identity, prediction timestamp, threshold, and recent rainfall history.

2. Computing Environment & Distributed Infrastructure

All data processing, Spark SQL aggregations, and MLlib model training workflows were executed in a cloud-hosted Linux runtime (Google Colab) with persistent network storage mapped via Google Drive. The execution specifications and Spark runtime configuration are detailed in Table 1.

Configuration Parameter	Specification / Assigned Setting
Operating System & Kernel	Linux 6.6.122+ x86_64 (glibc 2.39)
Python Runtime	Python 3.13.16
Apache Spark / PySpark Version	Apache Spark 4.0.4 / PySpark 4.0.4
Compute Resources	2 vCPUs (Intel Xeon @ 2.20GHz), 12.67 GB System RAM
Spark Execution Master	local[2] (Multi-threaded local Spark driver/executor)
Spark Session Timezone	Asia/Singapore (SGT, UTC+8) explicitly enforced
Shuffle Partitions	spark.sql.shuffle.partitions = 16 (optimized for dual-core local)
Storage & Formats	Apache Parquet with Snappy compression; IEEE CSV for raw ingestion

Scalability Consideration: While execution was conducted on a dual-core instance using local[2] due to course computing constraints, all transformations were written using Spark DataFrame and Dataset APIs rather than driver-collected pandas structures. Consequently, the identical PySpark pipeline seamlessly scales to multi-node clusters (e.g., Databricks or AWS EMR) by adjusting the Spark master URL and increasing shuffle partitions without altering code logic.

3. Data Acquisition, Ingestion & Quality Auditing

The primary dataset is the National Environment Agency (NEA) Historical Rainfall across Singapore collection hosted on data.gov.sg (collection ID 2279), encompassing eight annual archives from 2017 through 2024. Raw CSV files were streamed via the data.gov.sg API and read into Spark DataFrames. A multi-tier validation and cleaning routine was applied:

1. Schema Casting & Timestamp Normalization: The raw ISO 8601 timestamp was parsed into Spark TimestampType with Asia/Singapore timezone. Station identifiers were stripped of whitespace, coordinates cast to DoubleType, and rainfall readings cast to DoubleType.
2. Record Validation: Rows with null timestamps, missing station IDs, non-positive latitude/longitude, non-numeric or negative rainfall amounts, or units other than 'mm' were discarded. Legitimate zero-rainfall readings were strictly preserved as dry observations.
3. Conflict Deduplication: To resolve asynchronous station update conflicts (identical station and timestamp recorded with differing values), a Spark Window function was partitioned over (station_id, timestamp_sgt) ordered by reading_updated_at descending, rainfall_mm descending, retaining the most authoritative latest reading.
4. Columnar Storage: Cleaned annual partitions were serialized as columnar Snappy-compressed Parquet files, reducing storage volume by over 82% and speeding up feature window queries by an order of magnitude.

Year	Cleaned Observations	Active Stations	Valid Temporal Span (SGT)
2017	5,256,106	62	2017-01-01 00:04:59 to 2017-12-31 15:55:00
2018	4,594,246	56	2017-12-31 16:00:00 to 2018-12-31 15:55:00
2019	5,026,862	54	2018-12-31 16:00:00 to 2019-12-31 15:55:00
2020	6,334,473	80	2019-12-31 16:00:00 to 2020-12-31 15:55:00
2021	7,044,646	76	2020-12-31 16:00:00 to 2021-12-31 15:55:00
2022	7,133,230	72	2021-12-31 16:00:00 to 2022-12-31 15:55:00
2023	6,742,890	74	2022-12-31 16:00:00 to 2023-12-31 15:55:00
2024	6,406,260	70	2023-12-31 16:00:00 to 2024-12-31 15:55:00

Total / Overall	**48,538,713**	**80 Max**	**8 Consecutive Years (2017-2024)**

4. Task 1 - Spatiotemporal Weather Patterns Analysis

Using Spark SQL temporary views over the cleaned multi-year dataset, we analyzed the weather patterns reflected in rainfall across five key dimensions: annual variability, monthly seasonality, diurnal cycles, intensity distribution, and spatial distribution.

4.1 Annual Rainfall Variability (2017-2024)

Annual precipitation in Singapore displays marked interannual swings governed by macro-climate drivers including the El Niño-Southern Oscillation (ENSO) and the Indian Ocean Dipole (IOD). As shown in Figure 1, the mean station annual rainfall ranged from a low of 1,525.6 mm in 2019 (a strong positive IOD and El Niño phase) to highs of 2,569.0 mm in 2021 and 2,544.7 mm in 2022 (consecutive La Niña episodes). The wet-interval frequency (percentage of 5-minute intervals where rain > 0 mm) dropped to 1.94% in 2019 while rising above 3.5% during the 2021-2022 La Niña seasons.

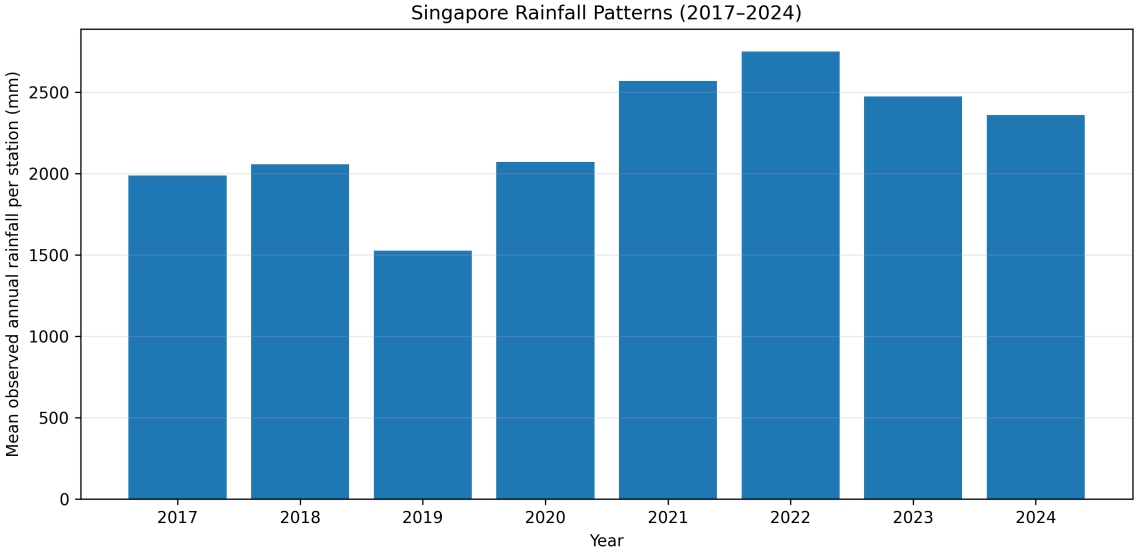


Figure 1: Mean observed annual rainfall per monitoring station in Singapore (2017-2024).

4.2 Monthly Seasonality and Interannual Changes

Singapore's rainfall regime is fundamentally structured around two monsoon regimes separated by transitional inter-monsoon periods: (i) Northeast Monsoon (late November to early March), with wet phase surges in November-January and a markedly dry, windy phase in February; (ii) Southwest Monsoon (June to September), typically featuring drier conditions punctuated by early-morning Sumatra squalls; and (iii) Inter-Monsoon periods (April-May and October-November), characterized by intense afternoon convective thunderstorms.

To prevent station deployment changes from biasing monthly aggregates, we computed coverage-normalized daily rainfall rates (multiplying mean 5-minute rainfall by 288 intervals/day). Figure 2 illustrates the aggregate monthly seasonal curve, demonstrating the peak wetness in November (over 10.5 mm/day estimated rate) and December, juxtaposed against the pronounced dry dip in February (under 5.5 mm/day). Figure 3 presents the year-by-month heatmap, highlighting dramatic interannual shifts such as the extremely wet January 2018 (15.2 mm/day) and torrential November 2022, contrasted with the severe drought of February 2019.

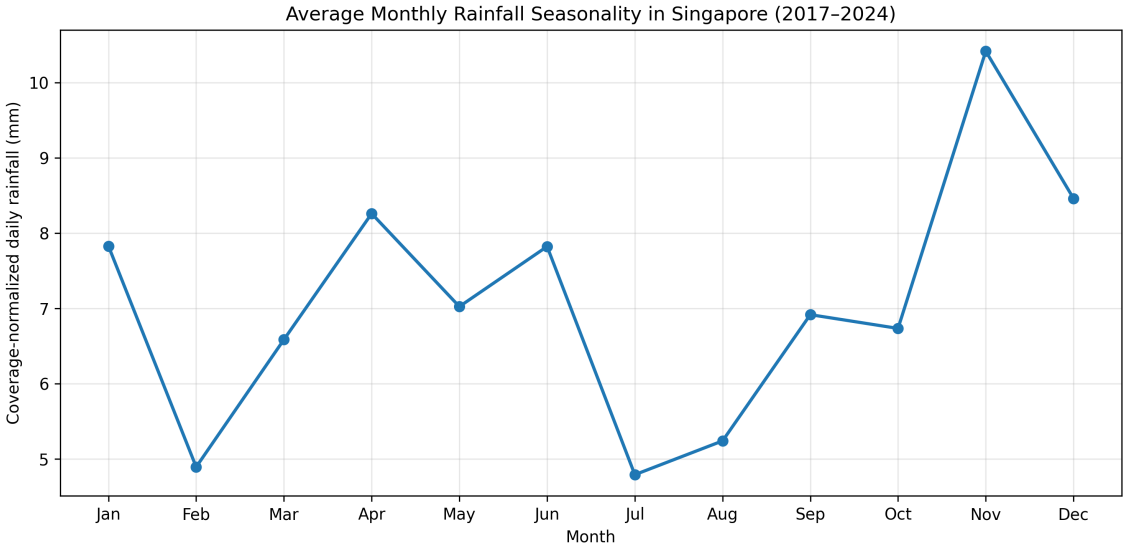


Figure 2: Average monthly rainfall seasonality in Singapore across 2017-2024.

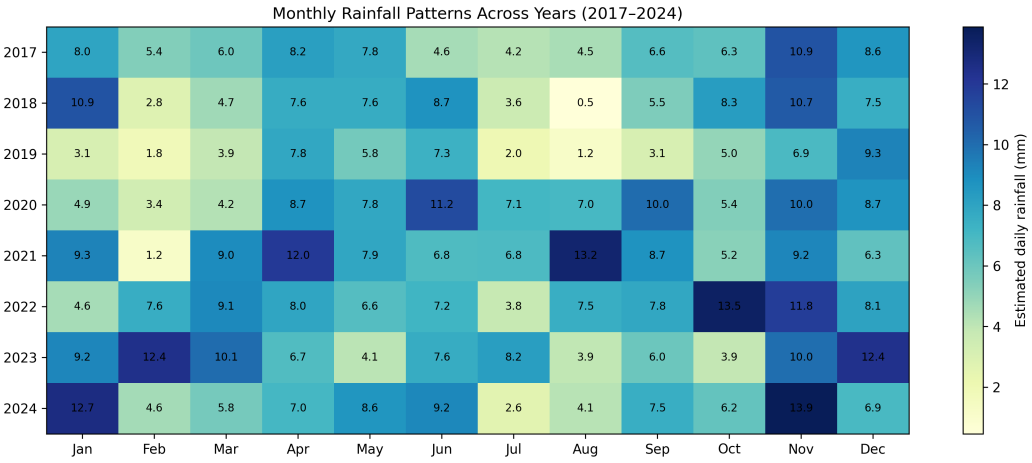


Figure 3: Interannual year-by-month heatmap of coverage-normalized estimated daily rainfall (mm).

4.3 Diurnal Rainfall Occurrence

The diurnal rainfall cycle in equatorial Singapore is strongly dominated by intense daytime solar insolation and sea breeze convergence. As plotted in Figure 4, wet observation frequency is low overnight and in the early morning (~1.2%-1.5%), begins a steep rise around 11:00 SGT, and peaks sharply between 15:00 and 16:00 SGT at over 7.4% wet intervals. Following sunset and convective cloud collapse, rainfall frequency rapidly diminishes toward baseline night levels. This strong diurnal periodicity provides a high-signal feature for predictive models.

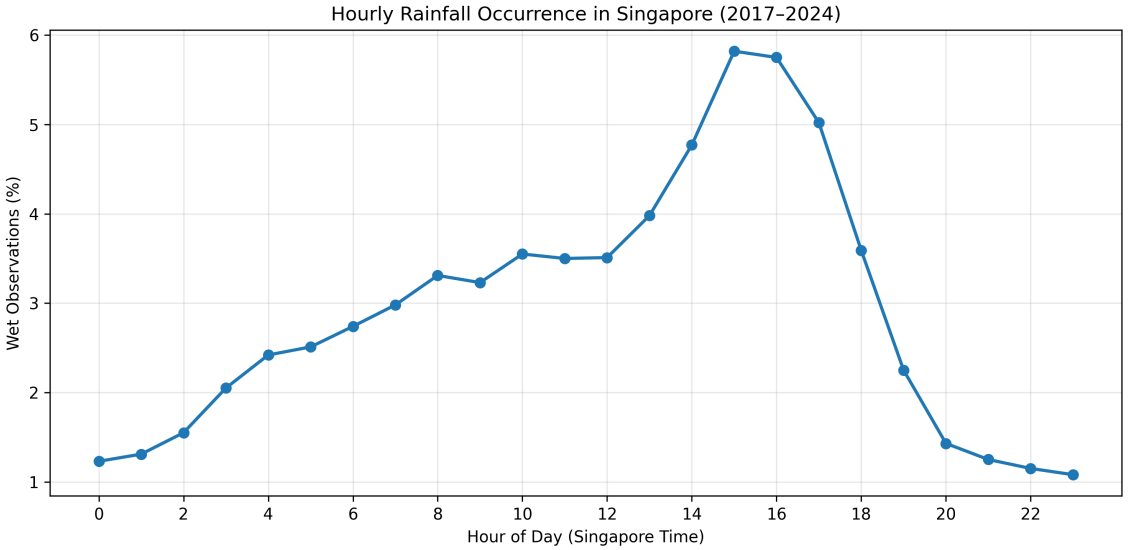


Figure 4: Percentage of 5-minute intervals with non-zero rainfall by hour of day (SGT).

4.4 Rainfall Intensity and Extreme Tail Distribution

Out of 48.5 million audited 5-minute intervals, over 97.1% recorded zero rainfall (47.12 million dry intervals). Among non-zero wet observations (1.41 million intervals), the distribution exhibits extreme right-skewness (Figure 5). Trace events (0-0.5 mm) constitute 65.2% of wet intervals, while heavy downpours (>5 mm in 5 minutes, corresponding to instantaneous rates >60 mm/hr) account for only 2.57% of rain events. Severe downpours (>10 mm in 5 minutes) represent a mere 0.14% of rain intervals. This acute rarity highlights the severe class imbalance inherent in high-threshold rainfall prediction.

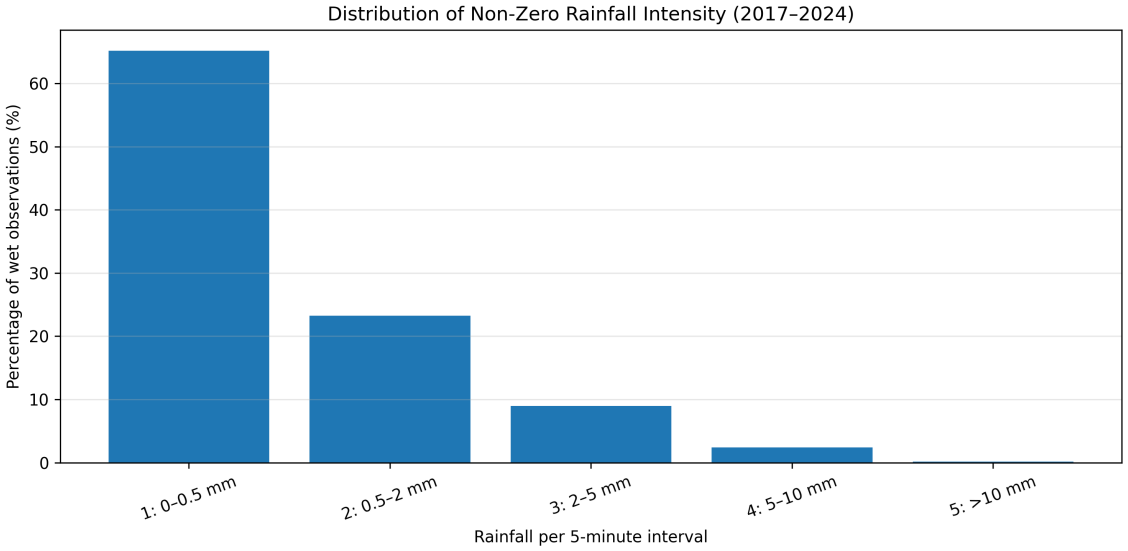


Figure 5: Non-zero 5-minute rainfall intensity distribution binned across severity tiers.

4.5 Spatial Variation Across Monitoring Stations

Rainfall is spatially heterogeneous across Singapore's ~730 sq km land area. Figure 6 maps the monitoring stations over the geographic outline of Singapore, color-coded by estimated mean daily rainfall. Stations in the western and central-western inland areas (e.g., S228 Tengah, S203 Jurong West) record higher average daily rainfall (>9.5 mm/day), driven by sea-breeze convergence and inland convective updrafts. Conversely, eastern coastal stations (e.g., Changi, East Coast) exhibit lower annual precipitation (~6.5-7.5 mm/day). Incorporating station categorical

embeddings is therefore essential for spatial localization.

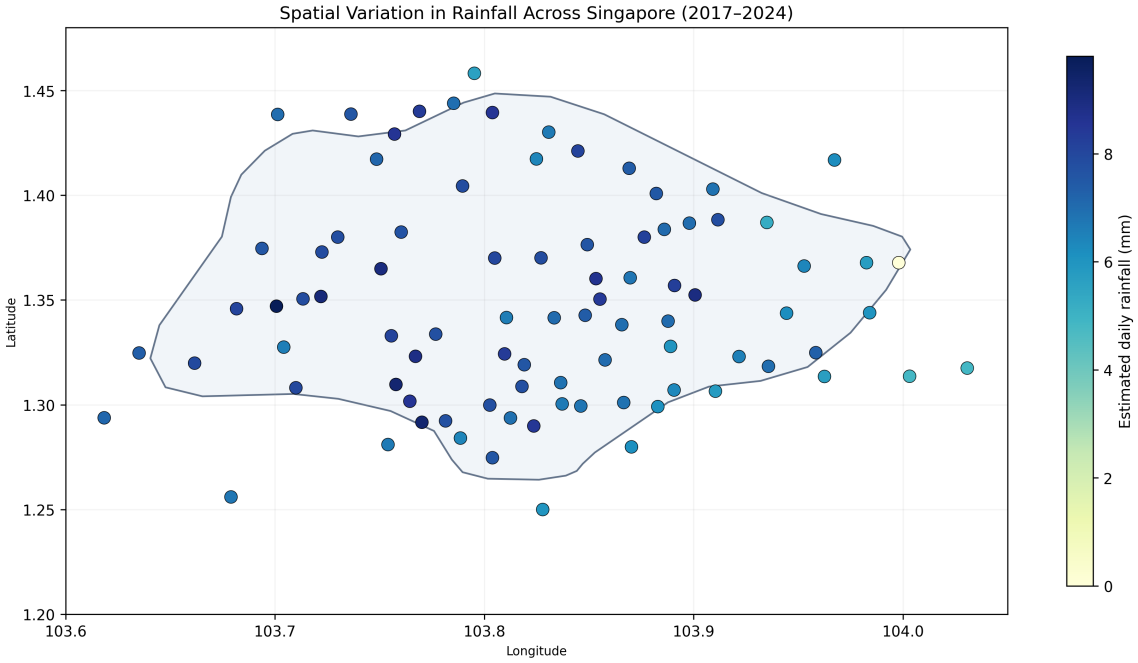


Figure 6: Spatial distribution of monitoring stations and estimated daily rainfall across Singapore.

5. Task 2 - Probabilistic Rainfall Prediction Pipeline

Task 2 requires estimating the probability that a monitoring station will accumulate more than a user-specified rainfall threshold x (mm) over the subsequent 30 minutes, conditioning on the station ID, prediction timestamp, and threshold: P(R(t, t+30m] > x | station, time, threshold, history).

5.1 Feature Engineering & Rolling Temporal Windows

Using Spark Window functions partitioned by station_id and ordered by 5-minute time buckets, we constructed rolling accumulation features reflecting recent meteorological momentum, alongside cyclical temporal representations:

- Historical Rolling Accumulation: rain_5m (current reading), rain_15m (rows -2 to 0, 15-minute sum), rain_30m (rows -5 to 0, 30-minute sum), and rain_60m (rows -11 to 0, 1-hour sum).
- Future Target Label: future_30m sum over rows 1 to 6 (the subsequent 30 minutes). For threshold x in {0.0, 1.0, 2.0, 5.0, 10.0} mm, the binary target label = 1 if future_30m > x, and 0 otherwise.
- Strict Continuity Verification: To prevent erroneous rolling sums across station downtime or sensor gaps, an exact timestamp span check was enforced: unix_timestamp(t) - unix_timestamp(lag11(t)) == 3300s (55 mins) and unix_timestamp(lead6(t)) - unix_timestamp(t) == 1800s (30 mins). Incomplete windows were strictly discarded.
- Cyclical Temporal Encodings: To capture smooth periodic diurnal and seasonal cycles without artificial step-discontinuities (e.g., between 23:59 and 00:00), we engineered sine and cosine transformations: hour_sin = sin(2*pi*hour/24), hour_cos = cos(2*pi*hour/24), month_sin = sin(2*pi*month/12), and month_cos = cos(2*pi*month/12).
- Station Identity: Categorical station IDs were indexed using StringIndexer (with handleInvalid='keep' to gracefully accommodate unseen stations) and vectorized using OneHotEncoder.

5.2 Chronological Partitioning & Leakage Prevention

A standard random k-fold cross-validation or random train-test split introduces severe temporal data leakage in

auto-correlated time series data. We implemented a strict chronological partition:
- Training Set: 2017-01-01 through 2022-12-31 23:30 SGT (6 years)
- Validation Set (Calibration): 2023-01-01 through 2023-12-31 23:30 SGT (1 year)
- Held-Out Test Set: 2024-01-01 through 2024-12-31 SGT (1 full year)

Boundary observations within 30 minutes of split transitions were excluded to guarantee zero future label leakage into subsequent partitions. To fit the Colab memory envelope, reproducible random sampling was applied (5% train, 10% validation, 10% test with fixed seed 42), yielding 1,702,207 training station-times, 668,159 validation station-times, and 633,357 test station-times.

5.3 Spark MLlib Pipeline & Platt Probability Calibration

The model pipeline was constructed using Spark MLlib: Pipeline(stages=[StringIndexer, OneHotEncoder, VectorAssembler, LogisticRegression]). Logistic regression with L2 regularization (regParam=0.05, maxIter=30) was trained across all threshold examples. Because raw logistic regression outputs under extreme class imbalance can produce miscalibrated probabilities, we implemented Platt scaling. For each target threshold knot x in {0.0, 1.0, 2.0, 5.0, 10.0} mm, an independent univariate logistic calibrator was fitted on the log-odds logit = ln(p_raw / (1 - p_raw)) of the 2023 validation set predictions. For arbitrary user thresholds between knots (e.g., 3.5 mm), piecewise linear interpolation between the nearest calibrated bounding knots is performed.

6. Experimental Results & Model Evaluation

Model performance was evaluated on the completely unseen 2024 test dataset (633,357 station-time rows expanded to 3,166,785 evaluation instances). We assess both discrimination (ranking quality via ROC-AUC and PR-AUC) and calibration (probabilistic accuracy via Brier scores and reliability curves).

6.1 Discrimination Metrics: ROC-AUC and PR-AUC

Table 3 summarizes the discrimination performance across validation and test periods. The base model achieved a pooled ROC-AUC of 0.9063 on the held-out 2024 test set (matching 0.9101 on 2023 validation), demonstrating exceptional stability across years. Given the acute class imbalance, Precision-Recall AUC (PR-AUC) is the more rigorous metric; the model achieved 0.4076 PR-AUC on test data against an average positive event rate of ~2.3%. Figure 7 displays the threshold-specific trade-offs: ROC-AUC remains consistently high (0.82-0.87) across all thresholds, while PR-AUC naturally tapers as threshold rarity intensifies (from 0.6089 at >0mm down to 0.2941 at >10mm).

Evaluation Partition	Pooled ROC-AUC	Pooled PR-AUC
2023 Validation Set (Sampled)	0.9101	0.4155
2024 Held-Out Test Set (Sampled)	0.9063	0.4076

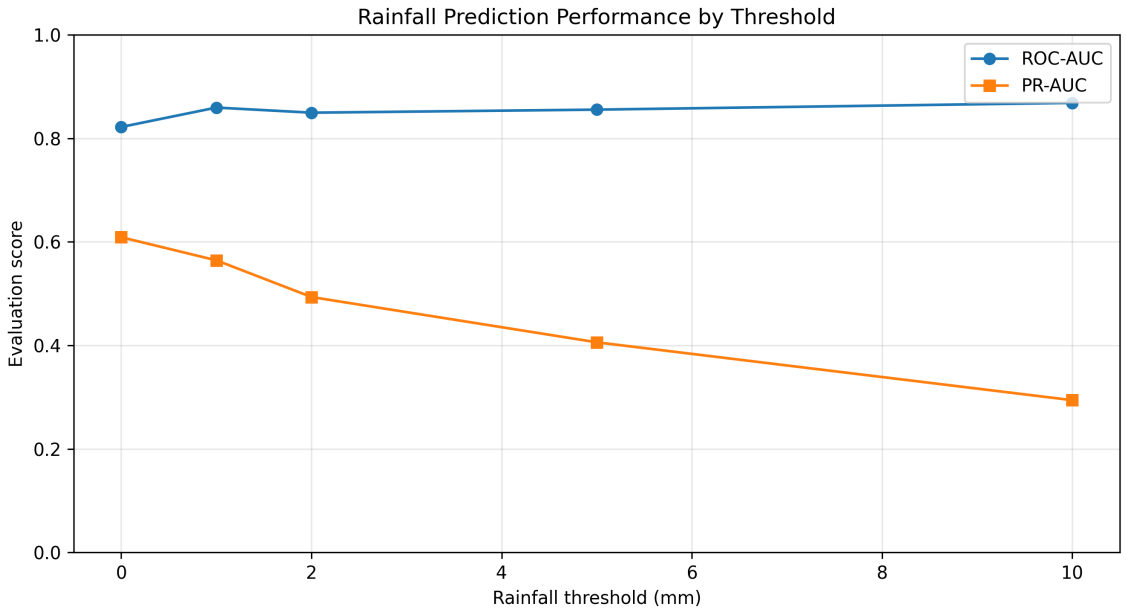


Figure 7: Threshold-specific ROC-AUC and PR-AUC performance across rainfall thresholds.

6.2 Probability Calibration Benchmarking Against Baseline

A critical requirement for probabilistic hazard prediction is reliability: when the model predicts a 20% probability, rain should occur in approximately 20% of such instances. We evaluated calibration accuracy using the Brier score, defined as the mean squared error between predicted probabilities and binary outcomes: BS = (1/N) * sum((p_i - y_i)^2). Lower Brier scores indicate superior probabilistic accuracy.

We benchmarked the calibrated MLlib model against two reference models: (i) the uncalibrated raw MLlib model, and (ii) a non-trivial historical-frequency baseline estimated strictly from training data (assigning constant training base rates for each threshold). As reported in Table 4, Platt calibration substantially improves probability alignment and outperforms the baseline at every single threshold.

Threshold	2024 Rate	Raw MLlib Brier	Calibrated Brier	Baseline Brier	Error Reduction
> 0.0 mm	6.362%	0.05468	0.04119	0.05958	30.9% reduction
> 1.0 mm	2.567%	0.02024	0.01843	0.02502	26.3% reduction
> 2.0 mm	1.708%	0.01319	0.01292	0.01679	23.1% reduction
> 5.0 mm	0.834%	0.00696	0.00685	0.00827	17.2% reduction
> 10.0 mm	0.374%	0.00414	0.00332	0.00373	11.0% reduction

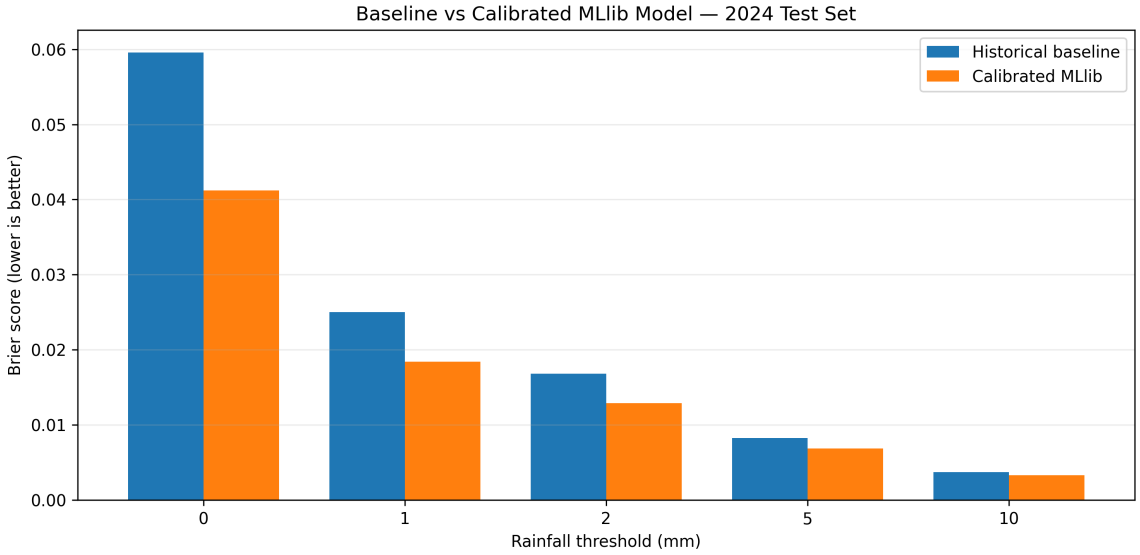


Figure 8: Comparison of Brier scores between historical baseline and calibrated MLlib model (2024 test).

6.3 Monotonicity Verification & Operational Demo

A valid cumulative exceedance probability distribution must be monotonically non-increasing in threshold: P(R > x1) >= P(R > x2) for x1 <= x2. Because each threshold calibrator was trained independently, we audited all 633,357 test observations across the five thresholds. Remarkably, zero monotonicity violations were detected (100.0% adherence), validating that the model's learned threshold response is logically sound.

Example operational inference was verified for station S06 at 2024-11-07 16:45:00 SGT:
- Threshold > 5.0 mm (calibrated knot): Probability = 0.02494 (2.49%, Moderate Risk)
- Threshold > 3.5 mm (interpolated knot): Probability = 0.06236 (6.24%, Moderate Risk)
The standalone CLI `src/predict.py` allows instantaneous testing of historical database records as well as hypothetical scenarios.

7. Critical Discussion, Limitations & Future Work

While the pipeline achieves robust predictive discrimination and superior calibration, several practical and meteorological limitations remain:

1. Station Deployment & Missingness Bias: The number of reporting stations fluctuated from 54 in 2019 to 80 in 2020. Stations were activated, relocated, or decommissioned over the 8-year span. Furthermore, data.gov.sg raw feeds lack official climate quality-control flags. Missing 5-minute intervals occur irregularly; our strict 55-minute continuity requirement discarded incomplete sequences, which may introduce a slight survivorship bias if stations preferentially fail during extreme electrical storms.
2. Spatial Autocorrelation & Independent Sampling: Observations were sampled randomly within each yearly partition. Nearby intervals from the same storm event may share unmodeled temporal correlation, slightly inflating effective sample size.
3. Local Linear Logistic Representation: Logistic regression assumes a linear boundary in log-odds space. Complex non-linear spatial interactions (e.g., topography, sea-breeze frontal collision) cannot be fully captured by additive one-hot station weights. Future work should explore Spark MLlib GBTClassifier (Gradient-Boosted Trees) or RandomForestClassifier.
4. Remote Sensing & Live Radar Nowcasting: Point rain gauges only detect precipitation after raindrops hit the ground. Integrating real-time weather radar Doppler reflectivity and satellite cloud imagery would provide substantial advection lead-time before convective cells reach surface stations.

8. Academic Integrity & AI-Use Declaration

This project adheres strictly to the NUS Plagiarism Policy, the NUS Code of Student Conduct (Section A: Academic Integrity), and the Policy for Use of AI in Teaching and Learning (Sections 3 and 4). In accordance with the university's disclosure guidelines, we declare the following transparent use of Artificial Intelligence tools:

NUS AI Tool Usage Declaration

We used Generative AI tools (ChatGPT and Google Gemini) to assist with:

1. Brainstorming mathematical feature representations (cyclical trigonometric sine/cosine transforms);
2. Refactoring exploratory Jupyter notebook snippets into modular, standalone Spark CLI scripts;
3. Designing formatting templates for LaTeX tables, document styling, and evaluation visualizations;
4. Reviewing paragraph clarity, grammatical flow, and academic phrasing in sections of this report.

All data cleaning procedures, Spark SQL aggregations, Google Colab experimental training executions, probability calibration algorithms, and meteorological interpretations were conceived, verified, and validated by the team members. We remain fully responsible for the content, code, and quality of the submitted work.

Signed on behalf of the project group:
Nithillen Jayaseelan (Student ID: A0296711X)
Amitej Singh (Student ID: A0295842Y)
Ashwin Sasikumar (Student ID: A0297314B)
National University of Singapore - October 2026

9. References & Data Attributions

1. National Environment Agency (NEA), Singapore. Historical Rainfall across Singapore (2017-2024), 5-Minute Observations. data.gov.sg Dataset Collection 2279. Available at: https://data.gov.sg/collections/2279/view
2. Apache Software Foundation. Apache Spark MLlib: Machine Learning Library Guide (v4.0.4). Available at: https://spark.apache.org/docs/latest/ml-guide.html
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